Sagar Singh Thapa

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Professional Summary

Computer Science undergraduate specializing in Data Science with hands-on experience developing machine-learning, GenAI, computer-vision, and predictive analytics applications. Experienced in data preprocessing, feature engineering, model evaluation, LLM integration, prompt engineering, and deploying ML workflows through FastAPI and Streamlit. Strong foundation in Python, SQL, Scikit-Learn, PyTorch, Transformers, and data-driven product development.

Technical Skills

AI/ML: Scikit-Learn, PyTorch, Transformers, Pandas, NumPy, Feature Engineering, Model Evaluation, Predictive Analytics, K-Means Clustering, Anomaly Detection, Computer Vision
GenAI: LangGraph, RAG, Hybrid RAG, Embeddings, LLM Integration, Prompt Engineering, Google Gemini API, Citation Verification
Languages & Data: Python, SQL, Java, C/C++, JavaScript, TypeScript, MongoDB Atlas, GeoTIFF, REST APIs, Data Pipelines
Backend & Development: FastAPI, Flask, Streamlit, React.js, Next.js, Tailwind CSS, Docker, Git, GitHub, Postman

Experience

IBM — Data Science Intern **Jun 2026 – Jul 2026**
Digital Advertising Analytics Project

- Collaborated in a three-member team to build an end-to-end digital advertising analytics platform covering data preparation, machine-learning workflows, backend APIs, frontend dashboards, and cloud-hosted data.
- Developed prediction, anomaly-detection, and K-Means clustering workflows to analyze advertising campaigns and generate actionable campaign segments and performance insights.
- Integrated ML outputs with the Google Gemini API to generate structured natural-language recommendations and served analytics workflows through FastAPI, Next.js, React, and MongoDB Atlas.

FYNTL AI, Sangrah Innovation Pvt. Ltd. — Web Development & AI Testing Intern **Jun 2025 – Aug 2025**
AI Analytics Product

- Tested AI-powered analytics APIs through structured Postman scenarios, validating response accuracy, reliability, edge cases, and overall system behavior against defined quality standards.
- Built responsive React.js and Tailwind CSS interfaces and debugged frontend/backend integration issues to improve product consistency and user experience.

IMPART (NGO) — Data Analytics Intern **Jun 2024 – Aug 2024**
Community Monitoring and Evaluation

- Cleaned and analyzed survey datasets using Python and SQL and prepared structured analytical reports to support community-focused monitoring, evaluation, and decision-making.

Projects

Multi-Agent AI Development Platform **Aug 2026 – Present**
Python | FastAPI | LangGraph | RAG | LLMs

- Building a multi-agent AI development platform with modular Research, Developer, and Platform subgraphs, integrated through a shared Main Graph and contract-driven service architecture.
- Developing the Research Agent and Hybrid RAG pipeline for intelligent document retrieval, embedding-based search, relevance ranking, and citation verification with structured confidence and retrieval scores.
- Implementing experiment planning and evaluation workflows using benchmark datasets, Precision@K, hallucination-rate analysis, and automated validation of retrieval and citation quality.
- Designing versioned API contracts and failure handling for reliable agent integration, including structured JSON logging, trace propagation, standardized error codes, and independently testable subsystem interfaces. “‘

Alpine Explorer **Aug 2026 – Present**
Multispectral Computer Vision | Swin Transformer | GeoTIFF | Python

- Building a geospatial AI platform using PlanetScope 8-band imagery, developing GeoTIFF preprocessing, UDM2/UDM2.1 quality masking, dataset construction, and semantic terrain segmentation using Swin Transformer.
- Developing reproducible ML experiments with GPU training, geographic train/validation/test splits, confidence estimation, and comparisons of RGB, multispectral, NDVI/EVI, and terrain features.
- Integrating terrain intelligence with route-risk analysis, offline journey tracking, cached geospatial data, and a grounded agentic AI copilot for evidence-based travel-risk explanations. “‘

Diabetic Tracker **Jan 2025 – Mar 2025**
Predictive Health Analytics | Python | Scikit-Learn

- Built a diabetes risk-prediction application covering dataset cleaning, missing-value handling, feature engineering, model training, and comparative model evaluation.
- Developed an interactive dashboard that accepts patient health attributes and displays model-generated risk predictions in an accessible, user-friendly format.

Education

University of Petroleum and Energy Studies (UPES), Dehradun **Aug 2023 – May 2027**
B.Tech in Computer Science & Engineering; CGPA: 7.3/10
Class XII (CBSE): 78%; Class X (CBSE): 82%